

COMPLETE AI DEV PIPELINE

ASSIGNMENT #10

ASSIGNMENT #10

COMPLETE AI DEV PIPELINE

STATUS: Mandatory

DUE DATE: 1 Sept 2026, 11:59 PM ET

STUDENT & GAME OVERVIEW

- **Student Name:** [Insert Your Name]
- **Capstone Game Title:** [Insert Game Title]
- **Game Concept Brief:** [Briefly describe your capstone game to verify the Pipeline-to-Game Connection]

DELIVERABLE 1: PLAYABLE LINK

▲ **GATE CRITERION:** A stranger must be able to open this link and play your game within 2 minutes without setup instructions. No link or a broken link results in a maximum 50% score across the entire assignment.

- **Playable Game Link:** [Insert URL here — e.g., Itch.io, Vercel, Netlify]

DELIVERABLE 2: PIPELINE SOURCE CODE & ENGINE INTEGRATION

- **Pipeline Repository Link:** [Insert GitHub/GitLab link]
- **Pipeline Run Video Link:** [Insert link to video evidence showing the pipeline running and generating the output assets/data]

INTEGRATION BREAKDOWN

- **Target Game Engine:** [e.g., Unity, Unreal, Godot, WebGL]
- **Automated Flow Description:** [Briefly explain how the agent output lands directly into the engine and functions in a scene without manual reformatting]

DELIVERABLE 3: PIPELINE AUDIT & COST ANALYSIS (1 PAGE)

01. Pipeline Production & Functionality

- a. What did the pipeline produce?** [List specific dialogue, assets, levels, etc., that are present in the playable build]
- b. What manual steps remain?** [Identify any manual friction points left in the pipeline]
- c. What would it take to eliminate them?** [State the solution required for 100% automation]

02. Architectural Reflection

- a. Current Architectural Decision to Change:** [Describe one choice you made that you would now do differently]
- b. Specific Alternative:** [Provide the exact alternative architectural approach you would implement]

03. Cost Analysis

- a. Total Actual Run Cost:** \$[Amount] (Must be calculated from the actual content generation run, not a hypothesis)
- b. Most Expensive Pipeline Step:** [Identify the specific step/prompt/model that consumed the most budget]
- c. Solo/Small-Team Sustainability:** [State whether this pipeline is financially sustainable for a small team and why]

04. Mid-Project Cost-Reduction Change

- a. **Strategy/Prompting Approach** — Before: [Describe original implementation] | After: [Describe optimized implementation]
- b. **Token / API Cost** — Before: [Insert original token count or cost] | After: [Insert new token count or cost]

GRADING RUBRIC (FOR REFERENCE)

CRITERION	DESCRIPTION	POINTS
Playable Link	A stranger can play within 2 mins without instructions. Gate: Broken/missing link = max 50% total score.	/ 2.0
Pipeline-to-Game Connection	Content in the game was traceably produced by the pipeline (verified via video/output). Anti-slop rule applies.	/ 3.0
Engine Integration	Output lands in the engine and functions without manual reformatting. Partial credit for 1 remaining documented manual step.	/ 2.0
Cost Analysis	Calculated against actual run. Identifies highest cost step and evaluates solo-dev sustainability.	/ 2.0
Pipeline Audit	1 page. Honestly details output, manual steps, 1 architectural change alternative, and a before/after mid-project cost reduction comparison.	/ 1.0

TOTAL SCORE: / 10.0

Grading note: If no playable link is submitted, the maximum score is 50% - the pipeline must produce a game someone can play